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PAPER CODE	U214-263-ESE
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(AY:2024-25) December 2024 (ENDSEM) EXAM
SY (SEMESTER - I)

COURSE NAME: SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Branch: CSE (Data Science) ✍ COURSE CODE: CD21233

SY (Pattern 2023)

Time: [1Hr 30 Min]

[Max. Marks: 40]

Instructions to candidates:

- 1) Figures to the right indicate full marks. Use of scientific calculator is allowed
- 2) Use suitable data wherever required
- 3) All questions are compulsory. Solve any two sub question each from Questions 1 and 2
- 4) Solve any one sub question (2 marks) from Questions 3 ,4 ,5 and 6 and sub question of 4 marks is compulsory from questions 3,4,5,and 6

Q. No.	Question Description	Max. Marks	CO mapped	BT Level
Q.1	a) In what ways does the Agile Process Model promote collaboration and communication more effectively than traditional process models like the Waterfall model?	[4]	1	Analysis
	b) How does the software process model selected for a project impact the overall quality and maintainability of the software product?	[4]	1	Analysis
	c) If a project has a high degree of uncertainty and the customer is unsure of the final product requirements, why might Agile or Evolutionary models be good choices than the Waterfall model?	[4]	1	Analysis
Q2	a) What are the primary goals of eliciting requirements, and how does this process help in forming a clear understanding of the system's needs from various stakeholders?	[4]	2	Understanding
	b) Why are non-functional requirements, such as performance and security, often more challenging to define and measure compared to functional requirements? What logical steps can be taken to ensure that non-functional requirements are captured accurately?	[4]	2	Understanding
	c) What is the purpose of a use case diagram, and how does it help in the system development process? Draw a use	[4]	2	Understanding

	case diagram for a Banking System.			
Q3	a) What is Agile design, and how does it differ from traditional software design methods? OR	[2]	3	Understanding
	b) What is the purpose of the design process in software development, and why is it important to follow a structured design approach?	[2]	3	Understanding
	c) How do you ensure the software's architecture remains scalable and maintainable while meeting the Agile principle of delivering working software quickly?	[4]	3	Analysis
Q4	a) Why is risk management an essential part of project management, and what are the main types of risks that can affect a software project? OR	[2]	4	Understanding
	b) Why is planning a software project crucial for project success, and what key elements should be considered during the planning phase?	[2]	4	Understanding
	c) You have a mix of experienced developers and junior developers in your team. How would you distribute tasks to ensure productivity while helping the juniors learn from the experienced team members?	[4]	4	Analysis
Q.5	a) How do PERT and Gantt charts assist in project scheduling and tracking? What is the difference in the way each chart represents project timelines? OR	[2]	5	Understanding
	b) What is the purpose of Configuration Management (CM) in a software project, and how do tools like version control systems help manage changes to project artifacts?	[2]	5	Understanding
	c) What logical trade-offs might the project manager face when deciding between using Function Point Analysis and Lines of Code (LOC) for estimating the size of this e-commerce project? How would these choices impact the allocation of development resources and potential risks in achieving scalability and performance?	[4]	5	Analysis
Q.6	a) What challenges might arise in managing Agile projects without proper traceability and visibility features like those offered in Azure DevOps? OR	[2]	6	Understanding
	b) How does the application lifecycle management (ALM) provided by Azure DevOps differ from traditional ALM processes?	[2]	6	Understanding

	C)How can Agile metrics in Azure DevOps, such as burn-down charts or cumulative flow diagrams, help in managing risks associated with the healthcare management system's delivery deadlines?	[4]	6	Analysis
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